

Bihar Engineering University, Patna
B.Tech. 3rd Semester Examination, 2023

Course: B.Tech.
Code: 101303

Subject: Computer-Aided Civil Engineering Drawing

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[2 x 7 = 14]**

- (a) What does the EXPLODE command do in AutoCAD?
 - (i) Breaks a complex object into its constituent parts.
 - (ii) Removes an object from a drawing.
 - (iii) Detaches an xref
 - (iv) Breaks a polyline into individual lines.
- (b) How do you create a 3D solid in AutoCAD?
 - (i) Using the solid command
 - (ii) Using the 3D command
 - (iii) Using the EXTRUDE command
 - (iv) Both a and c
- (c) The right-hand rule is used to determine the direction of the
 - (i) Positive X axis.
 - (ii) Positive Z axis.
 - (iii) Negative Y axis.
 - (iv) Negative X axis.
- (d) The ISOLINES system variable specifies
 - (i) The number of segments in a tabulated model
 - (ii) The number of contour lines that define a solid model.
 - (iii) The direction of contour lines in a solid model.
 - (iv) The length of the segments in an extrusion.
- (e) Which of the following is the correct statement for dimensioning a view?
 - (i) All of them should be placed within a view.
 - (ii) All of them should be placed close to the parts being dimensioned.
 - (iii) They should be marked from visible outlines rather than from hidden line.
 - (iv) Dimension indicated in one view may be repeated in another view.
- (f) A room of 3025 m² area is shown as 25 cm² area on a drawing sheet. What is the RF?
 - (i) 1/110
 - (ii) 1/1100
 - (iii) 1/220
 - (iv) 1/2200
- (g) Scale used for two systems of units measurement is
 - (i) Plain scale
 - (ii) Diagonal scale
 - (iii) Comparative scale
 - (iv) Vernier scale
- (h) Which line is drawn to make the section evident?
 - (i) Long break line
 - (ii) Chain thick
 - (iii) Border line
 - (iv) Hatching line
- (i) AutoCAD was first released in the year:
 - (i) 1980
 - (ii) 1982
 - (iii) 1984
 - (iv) 1986
- (j) How many units are available in AutoCAD?
 - (i) 4
 - (ii) 5
 - (iii) 6
 - (iv) 7

- Q.2** (a) Draw the detailed plan and section of RCC column and column footing with require details. **[7]**
- (i) Size of footing – 1200 mm × 1200 mm
 - (ii) Size of column – 300 mm × 300 mm
- (b) Write down the different super structure of building and explain the principles of planning of college building. **[7]**

- Q.3** (a) Give neat sketches and indicate conventional signs for the following: [7]
 (i) Railway line (ii) Timber (iii) Steel
 (iv) Plain concrete (v) Rock (vi) Plaster
 (vii) Brick Masonry
 (b) What is the difference between conventional sign & conventional symbol? [7]
 Explain with neat sketch.
- Q.4** (a) Define the followings: [7]
 (i) Filler / Extender (ii) Base (iii) Vehicle
 (iv) Thinner (v) Drier (vi) Pigment
 (b) Differentiate between gravity drainage system & siphonic drainage system [7]
- Q.5** (a) What do you understand by squint junction? Draw typical sketches showing squint junction in [7]
 (i) English bond (ii) Double Flemish bond
 (b) Define the following with neat sketch: [7]
 (i) Double Flemish bond (ii) English bond
 (iii) Racking bond (iv) Burnt clay perforated bricks
- Q.6** (a) It is proposed to construct a residential building (G+1) RCC framed structure with the following requirements: [7]
 (i) Drawing hall (ii) Living room
 (iii) Kitchen cum dining (iv) Master bedroom
 Provide veranda, passage, staircase, and sanitary unit as per requirement plot size of 50 m × 40 m. Draw floor plan for both levels.
 (b) What is BIM? Explain the component of Building Information Modeling (BIM). [7]
- Q.7** (a) How are the perspective projections classified? Mention application of each type of perspective projection. [7]
 (b) How do you begin a new drawing? How do you select an existing drawing for editing? List five important edit commands and mention their applications. [7]
- Q.8** (a) Explain the following: [7]
 (i) Reference plane (ii) Layers of AUTOCAD
 (iii) Optimal layout of drawing (iv) Picture plane
 (b) Explain the key points that must be considered for drawing the foundation plan of load-bearing structures. [7]
- Q.9** (a) Explain the various types of line used in civil engineering drawing and write down symbols are used in civil engineering drawing for electrical system? Draw neat sketches of electrical symbols. [7]
 (b) What are the benefits of AutoCAD from civil engineering point of view and explain reason for implementing AutoCAD. [7]

